

P1 · Problem-Framing Portfolio

First graded deliverable · CILO 1 · Week 2 · COMPLETED IN CLASS.

THE TASK

Four framed prediction problems — one per business function

Frame one business prediction problem for EACH of the four functional areas — marketing, financial, human resource, and operations — all grounded in **Kape Tayo Coffee**. Use the framing canvas on page 2 for every problem. No model, no code, no data work: P1 is judged entirely on the quality of your thinking before any analysis begins.

Your own track is your ANCHOR. For the area you own (P4 Anthonne · P5 Eliah · P6 Angus · P7 Lawrence), complete the extended anchor requirements on page 3 — this framing becomes the seed of your functional case report later in the term. The other three areas get the standard canvas. Start from your workshop C1 ladder if it held up to the shrug test.

How the Studio runs: your workshop C1 ladder becomes your ANCHOR canvas (30 minutes, full depth + memo). Then three rapid rounds of 15 minutes each — one canvas per remaining function, all twelve fields, one disciplined line per field. A 15-minute shrug-test gallery closes the Studio: classmates answer your prediction questions as the reader — act, or shrug. Precision beats polish; a rough honest canvas outscores a beautiful vague one.

Format: one page per problem (anchor may run to two). A document or markdown file in the repo; commit as you draft — history counts. Each submission carries an AI Use Log appendix (tool, date, purpose, what was produced, how verified). Tier 2 uses (brainstorming candidate problems, reviewing your own drafts) are welcome WITH disclosure; AI-drafted framing submitted as your own reasoning is Tier 4.

THE INSTRUMENT

Framing canvas — complete every field, in business language

Canvas field	What a complete answer contains
1 • Business decision	The recurring decision someone at Kape Tayo must make. A decision, not a topic: "how many trays of milk to order per branch per week," never "inventory."
2 • Decision-maker	The named reader who acts. What do they control? What can they actually change?
3 • Prediction question	One sentence, answerable in advance: "Which/how many/how likely ... in the next <horizon>?"
4 • Target variable	The exact thing predicted: how it is measured, in what units, over what window. If you cannot say how it would be measured, it is not a target yet.
5 • Unit of analysis	One row of your future dataset: a customer? an invoice? a branch-day? an employee-quarter? Be precise — most framing failures die here.
6 • Prediction horizon	How far ahead the prediction must arrive to still be useful to the decision-maker.
7 • Candidate predictors	What Kape Tayo would plausibly know BEFORE the outcome — name 4–8. Flag anything that smells like leakage (knowable only after the fact).
8 • Decision action	If the prediction says X, the decision-maker does Y. If no action changes, the prediction is decoration.
9 • Success criteria	In business units — pesos protected, hours saved, stockouts avoided, resignations pre-empted — with a number you would defend.
10 • Cost of error, both ways	What a false alarm costs vs what a miss costs, and which is worse here. Asymmetry is the interesting part.
11 • Baseline to beat	The dumb guess: recent average, majority class, existing rule of thumb. Your future model must beat this or it is not worth building.
12 • Scope & data ethics	Is this feasible with warehouse data this term? Any personal data (customers, employees)? If yes: RA 10173 applies — lawful basis, de-identification, and employee data is sensitive by default.

Reproduce these twelve fields for each of your four problems (a table, headings, or labelled paragraphs — your choice, all twelve present).

ANCHOR REQUIREMENTS

Extra depth for the track you own

For your own track only, add: (a) a one-paragraph memo TO your named reader, in their language, arguing why this prediction deserves their attention this quarter — open with the business pain, not the method; (b) two alternative prediction questions you considered and one sentence each on why you rejected them; (c) a line tracing the problem to something concrete — the workshop data, the announcement, or the Kape Tayo files ("I noticed ...").

CALIBRATION

The same problem, framed badly and framed well

Weak (would score Beginning):

"Predict sales using machine learning so management can make better decisions. Success is high accuracy."

Strong (what Exemplary looks like):

"Every Thursday the Operations Manager fixes each branch's weekend staffing and stock order (decision). Predict cups sold per branch-day (target; unit: branch-day) for the coming Fri–Sun (horizon), using day-of-week, payday proximity, local events, weather and the branch's trailing four weekends (predictors — all knowable Thursday). If predicted demand exceeds staffed capacity, add one floater barista and 15% stock (action). Success: stockout days cut from 6 to ≤ 2 per month across six branches without raising spoilage above 3% (criteria). A missed high day costs ~₱18k in lost sales and walkouts; a false alarm costs ~₱2.4k in idle wages — misses are ~7× worse, so we tolerate over-staffing (asymmetric cost). Must beat: same-weekend-last-month naive forecast (baseline). Branch-day sales data carries no personal data; nothing sensitive (ethics)."

GRADING

P1 rubric (counts in the laboratory / Problem-Framing component of your Prelim grade)

Criterion	Weight	Exemplary means
Target & unit precision	30%	Target measurable as stated; unit of analysis exact; horizon stated; no leakage among predictors.
Decision linkage	25%	A named decision-maker takes a stated action that changes with the prediction.
Success & cost of error	25%	Success in business units with a defensible number; both error directions costed; asymmetry recognised.
Baseline & feasibility	10%	Sensible dumb-guess baseline named; scope honest about this term's data.
Clarity & reader fit	10%	A business reader understands every sentence; anchor memo speaks the named reader's language.

House rules that apply: late policy –10%/calendar day (max 3); one re-review within 5 days with written basis; oral spot-checks — any canvas you cannot explain without notes is treated as unauthenticated. Framing has no "right answer": two of you may frame the same area differently and both score Exemplary. What is graded is precision, honesty about error, and fit to the reader.

Help available: your Week 2 bridging consultation doubles as a P1 feedback conversation — your rubric comes back there. Questions before Saturday: open an issue in the class repo.

WORKSHEET**Framing canvas — one sheet per problem**

Function: OPS

Anchor? Yes

Name: Lagdamen, Lawrence M.

Canvas field	Your answer — business language, one disciplined line minimum
1 • Business decision	What would our monthly inventory, staffing, operating capacity, and storage capacity for each branch look like in the next 6 months?
2 • Decision-maker	Operations Manager, who holds the power to adjust stocks, number of staff, and expected outcome per month.
3 • Prediction question	What is the ideal number of customer orders per day in a monthly timeframe?
4 • Target variable	Total number of orders per branch, per month.
5 • Unit of analysis	Branch-month; each row shows a monthly performance of each branch.
6 • Prediction horizon	6 months; it gives ample time for procurement and staffing problems.
7 • Candidate predictors	Historical monthly data for inventory, staffing numbers, and storage unit condition.
8 • Decision action	If demand is higher, increase stock and allocate more staff. If demand is lower, adjust stocks to an appropriate amount, and reduce staff onboarding.
9 • Success criteria	Reduce stockout by 15% per month, and storage spoilage or waste by 10%.
10 • Cost of error, both ways	Overprediction; this could cause overstaffing and spoilage of inventory.
11 • Baseline to beat	It should outperform a baseline of 3-month data.
12 • Scope & data ethics	Historical data is needed for the problem to be feasible.